

**B. TECH
(SEM V) THEORY EXAMINATION 2022-23
BIOINFORMATICS I**

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

- (a) What is meant by secondary database? What are the major secondary databases?
- (b) What is bioinformatics? What are the branches, scope and aim of bioinformatics?
- (c) What is the significance of E value in a BLAST result?
- (d) What are the features of PSI BLAST?
- (e) What do you understand by Affine Gap penalty?
- (f) Compare PAM & BLOSUM matrices.
- (g) Define Levenstein distance.
- (h) What is an outgroup? How to select one?
- (i) What do you mean by double blind approach in CASP?
- (j) What are the major interatomic forces that determine protein structure?

SECTION B

2. Attempt any three of the following:**10x3=30**

- (a) Give the applications of bioinformatics in drug discovery, QSAR, microbial genome and crop improvement.
- (b) Write down the significance of multiple sequence alignment in biological data analysis.
- (c) Discuss-
 - (i) Similarity matrix
 - (ii) Functional site detection in DNA
- (d) What are the methods of analyzing Phylogenetic trees? Explain.
- (e) Justify that how bioinformatics is helping in novel drug designing? Explain with suitable example.

SECTION C

3. Attempt any one part of the following:**10 x 1 = 10**

- (a) Explain the common bioinformatics pipeline in identifying the homologues gene regions in thesequence data of an unknown prokaryotic organism.
- (b) What is Entrez? Schematically represent the architecture of Entrez System, briefly explaining each of them.

4. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Explain the working of BLAST based on your knowledge of sequence alignment.
- (b) Discuss the difference between local and global alignment with suitable examples.

5. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Explain the concept of scoring matrices for aligning amino acid sequences. Briefly explain how PAM is derived?
- (b) There are two given sequences- S1 = TATTCGCAAG and S2 = GCTATATCCA. Obtain the optimal global alignment using dynamic programming method. Use any Scoring scheme of your choice.

6. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Define a Root, Node and Clade. How the distances between species are calculated for phylogenetic tree construction? Explain.
- (b) List out the steps involved in phylogenetic tree construction and discuss with a distance-based method.

7. Attempt any *one* part of the following: 10 x 1 = 10

- (a) What is the rationale behind homology modelling? Discuss the various steps involved in the process.
- (b) Write about-
 - (i) Motifs
 - (ii) RASMOL